PROJECT REPORT

E-Com Minutes: An AI-Powered Multi-Tenancy Automated E-Commerce Website Generator

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BONAFIDE CERTIFICATE

Certified that this project report titled "E-Com Minutes: An AI-Powered Multi-Tenancy Automated E-Commerce Website Generator" is the bonafide work of Akash Mitra, Shimanta Guha, Ritam Bera, and Jotirmoy Masanta, who carried out the project work under my supervision.

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ABSTRACT

E-Com Minutes is an AI-powered, single-codebase multi-tenancy platform that automatically generates fully functional, customized e-commerce websites for small and medium product-based businesses. The platform addresses the critical digital gap where 99% of Indian MSMEs lack an online presence due to high costs, technical complexity, and long development cycles.

The system combines three core components: an AI-powered setup wizard that generates complete websites in under 30 minutes, a fully functional e-commerce storefront with product catalog, cart, checkout, and order tracking, and a comprehensive admin dashboard with dynamic customization capabilities. The platform follows a single-codebase multi-tenancy model where one codebase serves unlimited clients with all customizations stored in the database.

The technical architecture uses React.js with Vite for the frontend, Supabase (PostgreSQL) with Row Level Security for backend database and authentication, and integrates Google Gemini API for setup automation and SmolLM2 for the AI sales assistant chatbot. The platform is designed for scalability, performance, and security with hidden admin access and complete data isolation between tenants.

Through a phased development approach spanning four months, E-Com Minutes demonstrates practical applications of full-stack web development, multi-tenancy architecture, AI integration, and automated deployment strategies.

Keywords: E-Commerce, Multi-Tenancy, Artificial Intelligence, Single-Codebase, Automated Deployment, Dynamic Customization

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Background

The digital economy has transformed how businesses operate, with e-commerce becoming essential for survival and growth. However, small and medium businesses face significant barriers when establishing an online presence. Custom development is expensive, technical platforms require expertise, and subscription services create ongoing financial burdens.

1.2 Problem Statement

Product-based MSMEs face critical barriers to e-commerce adoption:

Barrier	Description
Digital Gap	99% of Indian MSMEs (≈4.7 crore businesses) lack a functional website
Cost	Custom development costs ₹50,000 – ₹5,00,000
Time	Custom builds take 16–28 weeks
Recurring Fees	Platforms like Shopify charge ₹3,000–₹35,000/month
Technical Expertise	Platforms like WooCommerce require ongoing developer support
Customization	Affordable solutions offer limited branding flexibility

Table 1.1 — Barriers to e-commerce adoption faced by MSMEs

1.3 Proposed Solution

E-Com Minutes addresses these challenges through an AI-powered, single-codebase multi-tenancy platform that automatically generates customized e-commerce websites. The solution combines affordability, simplicity, complete customization, rapid deployment, self-service capabilities, and robust security.

2. OBJECTIVES

2.1 Primary Objectives

#	Objective	Description
1	Single-Codebase Multi-Tenancy	Develop a platform serving unlimited clients with complete data isolation
2	E-Commerce Storefront	Build product catalog, cart, checkout, payment, and order tracking
3	Admin Dashboard	Create comprehensive management with dynamic customization
4	Dynamic Customization	Enable logo, colors, fonts, and branding updates without code
5	AI-Powered Automation	Automate website generation, product description, and customer support
6	Security	Implement hidden admin access and row-level security for data isolation

Table 2.1 — Primary objectives of the project

2.2 Secondary Objectives

#	Objective	Description
7	Performance Optimization	Achieve sub-2.5s page load and sub-100ms API response
8	Scalability	Handle growing product catalogs and client bases
9	Documentation	Provide comprehensive technical and user documentation

Table 2.2 — Secondary objectives of the project

3. LITERATURE SURVEY

3.1 E-Commerce Platforms

The e-commerce platform market has evolved significantly over the past decade. Major platforms include Shopify, WooCommerce, and Magento, each with distinct limitations. Shopify operates on a software-as-a-service model with monthly fees and transaction charges. WooCommerce requires WordPress expertise and separate hosting management. Magento is resource-intensive and demands significant technical expertise.

3.2 Multi-Tenancy Architecture

Multi-tenancy has emerged as the preferred architecture for software-as-a-service platforms. Research demonstrates that multi-tenant architectures reduce development costs by 60-80% compared to single-tenant solutions. The single-codebase approach ensures consistency across all clients, simplifies maintenance, and enables rapid onboarding of new customers.

3.3 Database-Driven Customization

Database-driven configuration is a proven approach for scalable web applications. Studies show that storing all configuration in databases reduces deployment time by 80% compared to code-based configurations. This approach enables dynamic customization without requiring code changes or redeployment.

3.4 Supabase for Web Applications

Supabase has emerged as a leading open-source backend platform built on PostgreSQL. It provides database services, authentication, storage, and real-time capabilities. For e-commerce applications, Supabase offers significant advantages including row-level security for data isolation, scalability for handling large product catalogs, and built-in authentication services.

3.5 Artificial Intelligence in E-Commerce

AI integration has become increasingly important in e-commerce. Recommendation systems drive significant revenue for platforms like Amazon. AI-powered chatbots reduce customer support response times by 80%. Automated onboarding systems reduce setup time from weeks to minutes. Recent studies demonstrate that large language models can effectively automate complex business processes including website configuration and product description generation.

3.6 Security Considerations

Security remains paramount in e-commerce applications. Industry standards include PCI-DSS compliance for payment handling, row-level security for data isolation, JWT tokens for stateless authentication, and bcrypt hashing for password storage. The Open Web Application Security Project recommends returning

404 status codes for unauthorized admin access attempts to prevent attackers from discovering admin interfaces.

4. METHODOLOGY

The project follows a phased development approach over four months, with each phase building upon the previous.

4.1 Phase 1: Core E-Commerce Website Development (Weeks 1-4)

4.1.1 Technology Selection

Layer	Technology	Purpose
Frontend	React.js + Vite	Fast builds, component-based UI development
Styling	Tailwind CSS + Framer Motion	Responsive design, smooth animations
State Management	React Context API	Global state without external dependencies
Backend	Supabase (PostgreSQL)	Database, Auth, Storage, Realtime
Security	Row Level Security (RLS)	Multi-tenant data isolation
Payments	Razorpay	Indian payment gateway integration
AI	Google Gemini API	Setup automation, content generation
AI Chatbot	SmolLM2 (Hugging Face)	Sales assistant chatbot
Deployment	Vercel + Supabase Cloud	Automated hosting and scaling

Table 4.1 — Technology stack selected for Phase 1

4.1.2 Database Schema Design

The database schema includes tables for:

Table	Purpose	Key Fields
tenants	Multi-tenant isolation	id, name, subdomain, settings (JSONB)
products	Product catalog	id, tenant_id, name, description, price, category, stock, image_url
orders	Order management	id, tenant_id, order_number, user_id, total, status, shipping_address
order_items	Order details	id, order_id, product_id, quantity, price_at_time
users	Authentication	id, tenant_id, email, password_hash, role
shop_settings	Dynamic customization	tenant_id, logo_url, primary_color, secondary_color, font_family

Table 4.2 — Core database schema

Row Level Security (RLS): Each tenant's data is isolated using tenant_id policies. Customers only see their own orders. Admins only access their store's data.

4.1.3 Storefront Development

- Homepage with hero banners, categories, and trending products
- Shop page with filtering and sorting capabilities
- Product details pages with images, descriptions, and pricing
- Shopping cart using localStorage for persistence
- Checkout process with address collection and payment processing
- Order tracking with order number and email

4.1.4 Payment Integration

Razorpay integration for secure payment processing:

- Razorpay Checkout API integration
- Support for Credit/Debit Cards, UPI, NetBanking, Wallets
- Webhook handling for automatic order status updates

4.2 Phase 2: Admin Dashboard Development (Weeks 5-8)

4.2.1 Hidden Admin Access

- Admin panel accessible only through direct URL (/admin/login)
- No public links to admin page anywhere on the site
- Unauthorized access attempts return 404 status codes
- Authentication using Supabase Auth with JWT validation

4.2.2 Dashboard Features

Feature	Description
Dashboard Home	Key metrics: orders, revenue, products, customers
Product Management	Add, edit, delete, import, export products
Order Management	View, update status, add tracking information
Dynamic Customization	Logo, colors, fonts, contact details management
Analytics	Revenue charts, order statistics, top products
Promotions	Hero banners, discount coupons

Table 4.3 — Admin dashboard features

4.2.3 Dynamic Customization Module

The customization module enables business owners to:

Setting	Description
Shop Name	Update store name displayed everywhere
Logo Upload	Upload PNG, JPG, SVG (max 2MB) to Supabase Storage
Favicon	Upload website icon
Primary Color	Color picker for main brand color
Secondary Color	Secondary brand color
Accent Color	Accent/button color
Font Family	Choose from curated font list
Contact Details	Email, phone, address
Social Links	Facebook, Instagram, Twitter, YouTube

Table 4.4 — Dynamic customization settings

All customizations are stored in Supabase and applied dynamically to the storefront.

4.3 Phase 3: AI Integration and Automation (Weeks 9-12)

4.3.1 AI-Powered Setup Wizard

The setup wizard automates website generation through:

Step	Description	AI Role
1. Brand Identity	Shop name, logo upload, color selection	AI suggests color palette from logo
2. Product Import	CSV upload or manual entry	AI generates product descriptions
3. Admin Account	Email and password creation	Supabase Auth handles creation
4. Payment & Shipping	Gateway selection, shipping zones	Configuration
5. Deploy	Review and one-click deployment	Automated deployment

Table 4.5 — AI-powered setup wizard steps

AI Functions:

Function	Input	Output
Store Configuration	Brand name, logo, colors	JSON configuration
Product Description	Product name, category	SEO-optimized description
Color Scheme	Logo image	Complementary color palette
SEO Metadata	Shop name, products	Meta titles, descriptions

Table 4.6 — AI functions used in the setup wizard

4.3.2 AI Sales Assistant Chatbot

Feature	Description
Model	SmolLM2 (360M parameters, lightweight)
Deployment	Hugging Face Spaces (free tier)
Constraint	Only discusses products available in the store
Function	Product recommendations and customer questions
Context	Maintains conversation history

Table 4.7 — AI sales assistant chatbot specification

4.3.3 Automated Deployment

- Environment configuration
- SSL certificate provisioning
- Post-deployment verification
- One-click deployment capability

4.4 Phase 4: Testing and Documentation (Weeks 13-16)

4.4.1 Testing Procedures

Test Type	Description
Unit Testing	Validate React components and API endpoints
Integration Testing	Verify checkout flow and admin operations
Performance Testing	LCP <2.5s, API <100ms
Security Testing	Admin access protection, JWT validation, input sanitization
RLS Testing	Verify tenant data isolation

Table 4.8 — Testing procedures

4.4.2 Documentation

Document	Purpose
Technical Documentation	System architecture, database schema, API references
User Documentation	Admin panel guide, setup instructions
Deployment Guide	Hosting configuration instructions

Table 4.9 — Documentation deliverables

5. PROJECT TIMELINE

Phase	Weeks	Activities
Phase 1	1-4	Technology selection, database design, authentication, storefront development, cart and checkout
Phase 2	5-8	Admin dashboard, hidden authentication, product and order management, dynamic customization
Phase 3	9-12	AI setup wizard, sales assistant chatbot, automated deployment
Phase 4	13-16	Comprehensive testing, performance optimization, documentation, final polish

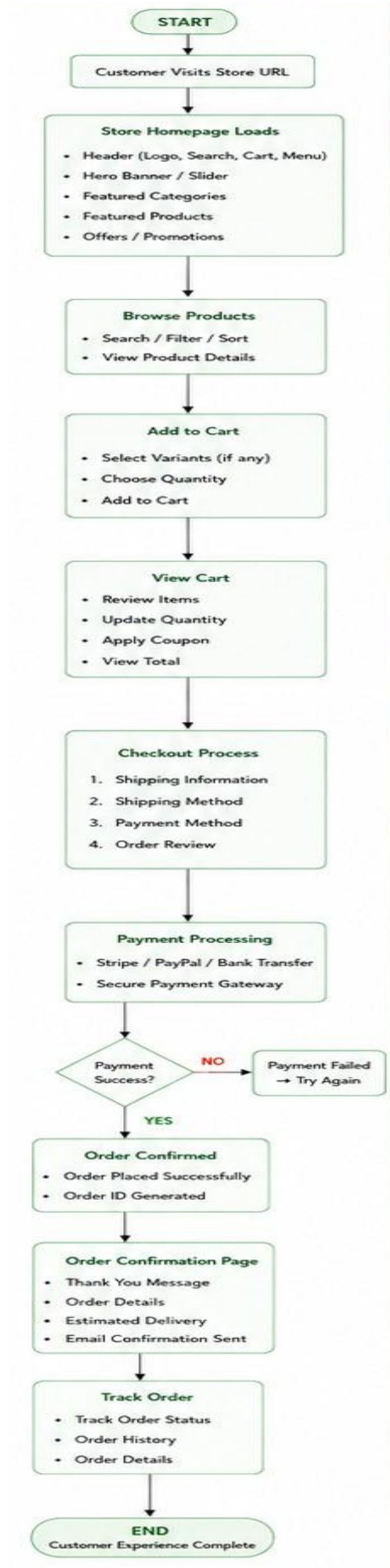
Table 5.1 — Four-month phased project timeline

6. TECHNICAL STACK

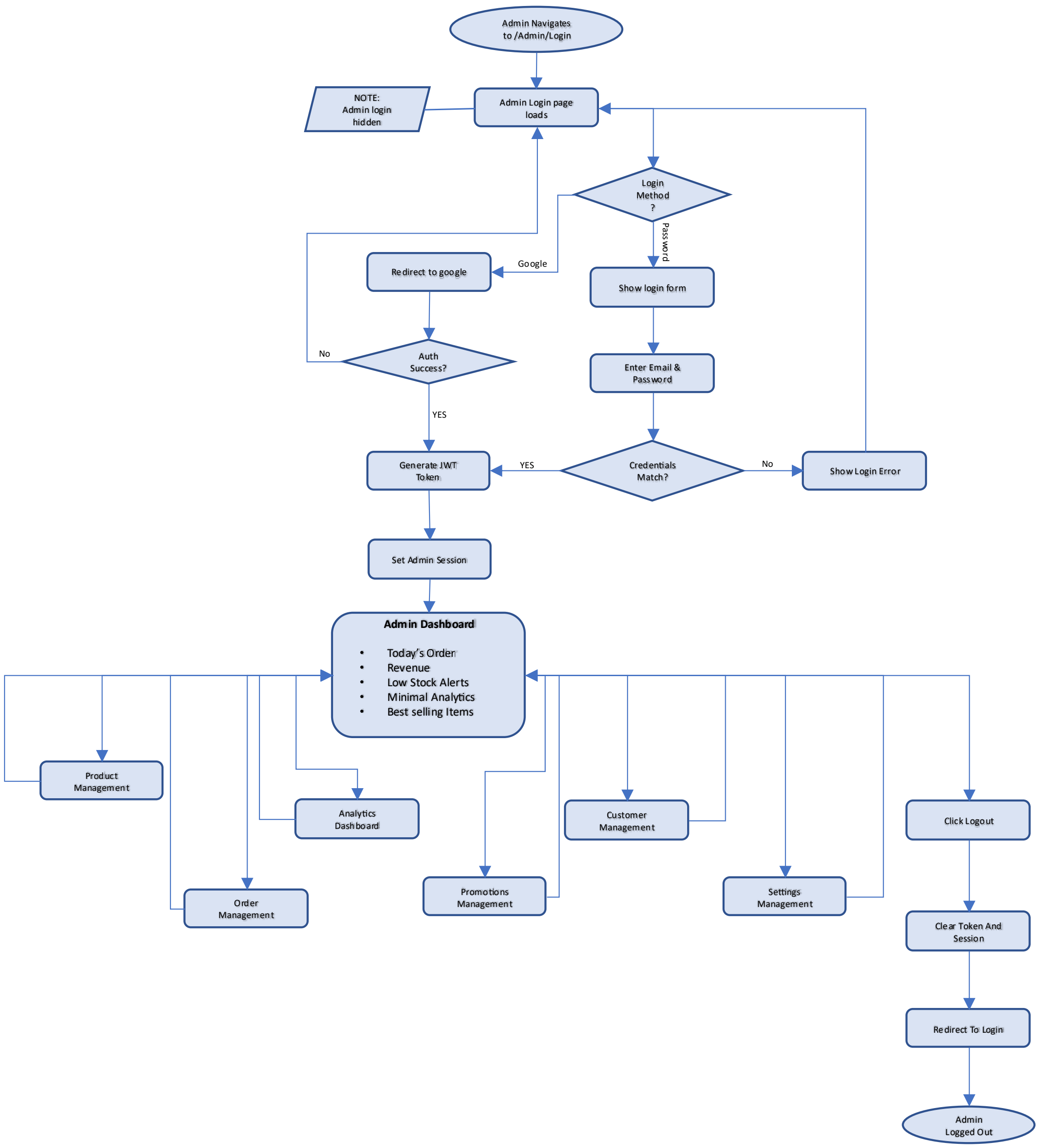
Component	Technology	Purpose
Frontend Framework	React.js + Vite	Component-based UI development
Styling	Tailwind CSS + Framer Motion	Responsive design, animations
State Management	React Context API	Global state management
Backend	Supabase (PostgreSQL)	Database, Auth, Storage, Realtime
Security	Row Level Security (RLS)	Multi-tenant data isolation
Authentication	Supabase Auth (JWT)	Secure user authentication
File Storage	Supabase Storage	Product images, logos
Payment	Razorpay	Indian payment gateway
AI Setup	Google Gemini API	Website generation automation
AI Chatbot	SmolLM2 (Hugging Face)	Customer support assistant
Deployment	Vercel + Supabase Cloud	CI/CD, scaling, hosting

Table 6.1 — Complete technical stack

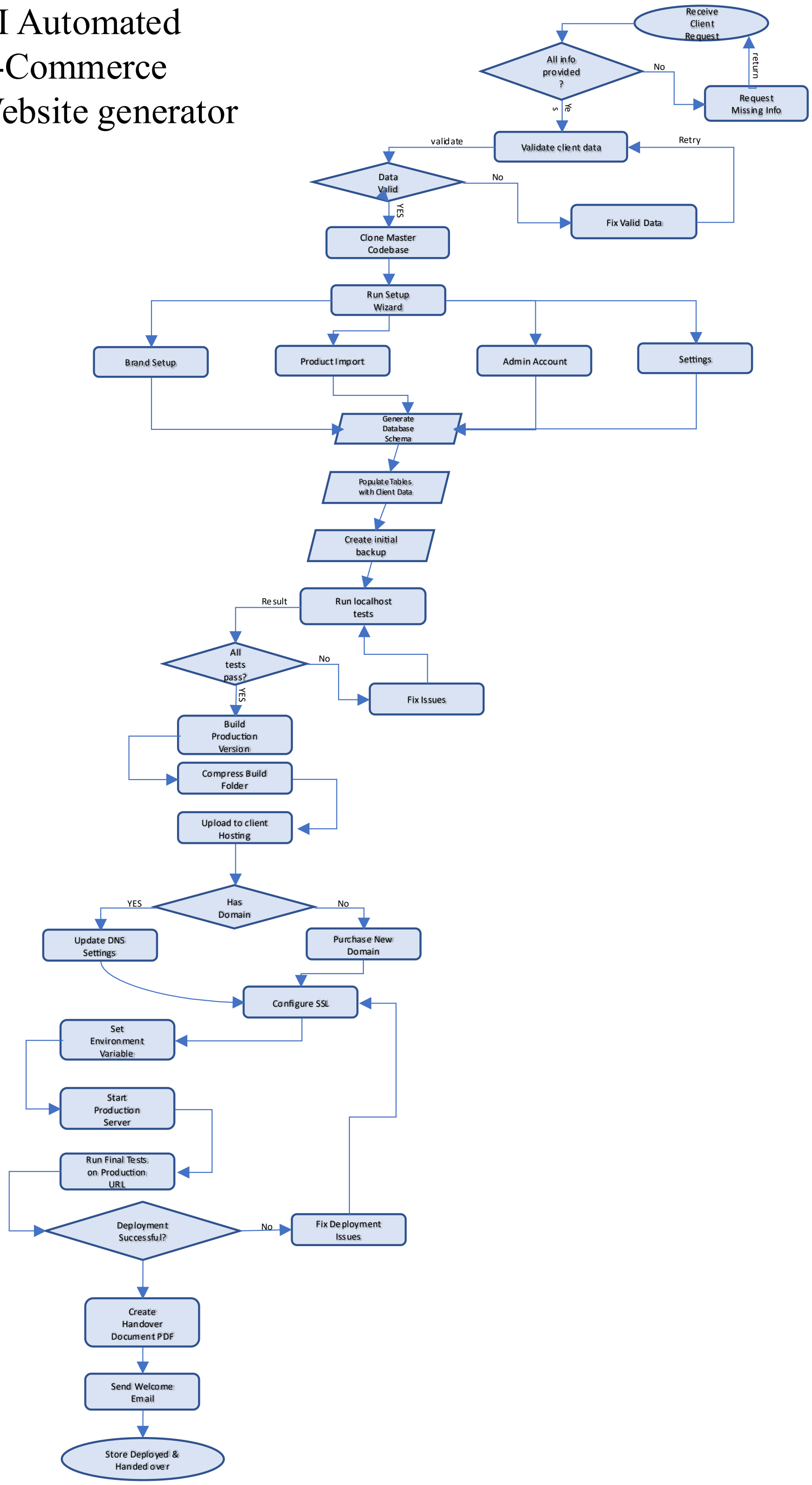
E-Com
Website
Workflow



Admin Dashboard Workflow



AI Automated E-Commerce Website generator



8. EXPECTED OUTCOMES

Outcome	Metric	Technical Validation
Website Deployment Time	<30 minutes	Automated setup wizard
Page Load Time (LCP)	<2.5s	React + Vite optimization
API Response Time	<100ms	Supabase + RLS indexing
Infrastructure Cost	30-50% lower vs single-tenant	Multi-tenancy architecture
AI Sales Assistant	Real-time responses	SmolLM2 implementation
Multi-Tenancy	Unlimited clients	RLS-based isolation
Dynamic Customization	Complete branding control	Database-driven configuration
Admin Security	Hidden login, 404 on unauthorized	Security architecture

Table 8.1 — Expected outcomes and validation metrics

9. CONCLUSION

E-Com Minutes addresses the critical digital gap where 99% of Indian MSMEs lack a functional website. By combining AI-powered automation with single-codebase multi-tenancy architecture, the platform:

1. Reduces deployment time from 16-28 weeks to under 30 minutes through automated setup wizard
2. Cuts infrastructure costs by 30-50% through multi-tenancy architecture
3. Eliminates recurring platform fees with a one-time cost model
4. Provides complete branding control through dynamic customization without code changes
5. Demonstrates technical excellence in full-stack development, multi-tenancy, AI integration, and security

The project showcases practical applications of:

- Multi-tenancy architecture with Row Level Security for data isolation
- AI-powered automation using Gemini API and SmolLM2
- Modern full-stack development with React, Supabase, and PostgreSQL
- Security-first design with hidden admin access and JWT authentication

Future enhancements could include multi-language support, mobile applications, advanced AI recommendation systems, and visual search capabilities.

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